power

File: power.kicad_sch

pi_header

File: pi_header.kicad_sch

radio

File: radio.kicad_sch

ui

File: ui.kicad_sch

Generated by hw/scripts/schgen.py
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**MTP F26 Team C (UPC)**

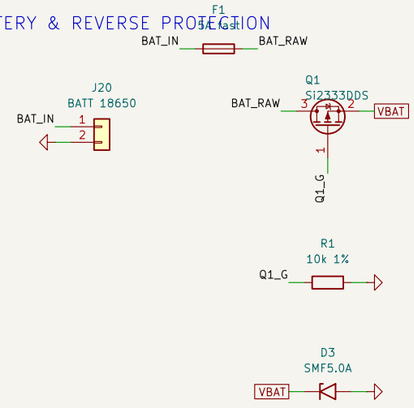
Sheet: /
File: teamc-carrier.kicad_sch

***Title: Team C Pi 3 A+ HAT carrier (DRAFT - for ET review)***

| Size: A4 | Date: 2026-10-19 | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/5 |

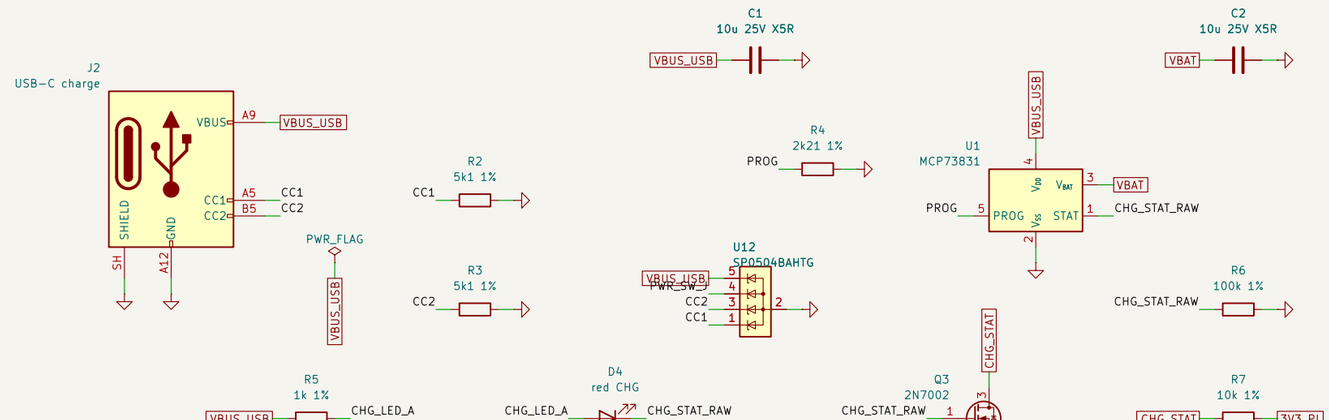
# POWER - one protected 18650 → 5.1 V for the Pi 3 A+ and 3.3 V for the radio (DRAFT - for ET review)

## 1 BATTERY & REVERSE PROTECTION



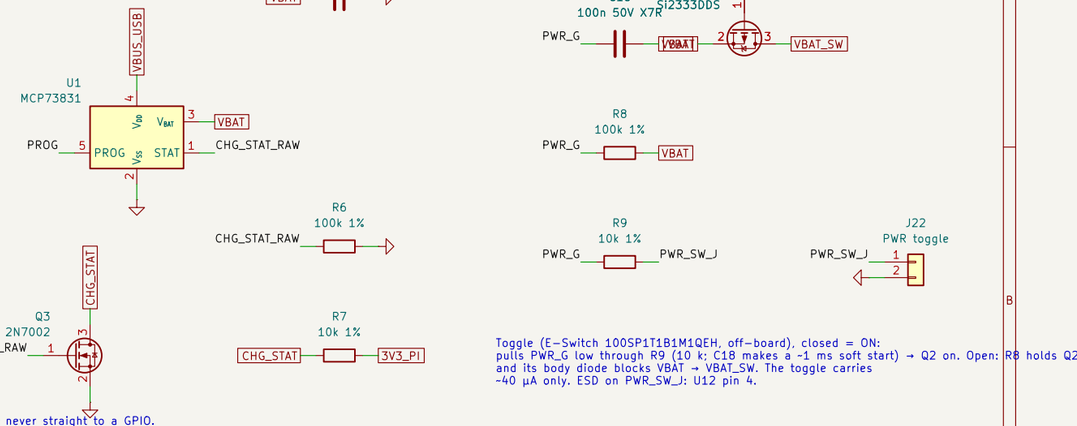

Cells: protected 18650 (PCM inside) in an MPD BH-18650-PC
holder on the box floor (off-board), lead to J20 (JST-VH, 10 A).
2nd cell: a 2nd holder wired in parallel into the same J20 lead (harness; matched cells only).
Q1: the body diode conducts for a correct cell, then VGS = −VBAT
turns it on; a reversed cell is blocked. D3: ESD/hot-plug clamp.
Do not plug USB with a reversed cell (the charger, current-limited,
would bias Q1 partly on).

## 2 CHARGER & USB-C (charge-only, 452 mA; charges with the switch OFF)




CC1/CC2 5.1 k = sink (default USB 5 V). RPROG 2.21 k → 452 mA
(1000 V / RPROG). VC of the SMF5.0A (9.2 V) > MCP73831 VDD abs. max
7 V: ESD only, not a guard against a faulty charger. No power path:
the charger feeds VBAT, upstream of the power switch.

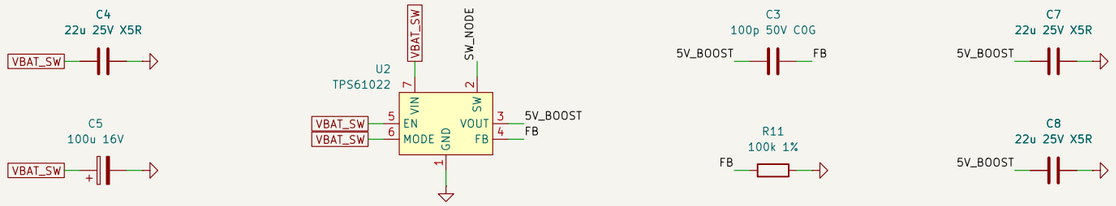
## 3 POWER SWITCH (P-FET, panel toggle on J22)




Toggle (E-Switch 100SP1T1B1M1QEH, off-board), closed = ON:
pulls PWR_G low through R9 (10 k; C18 makes a ~1 ms soft start) → Q2 on. Open: R8 holds Q2
and its body diode blocks VBAT → VBAT_SW. The toggle carries
~40 µA only. ESD on PWR_SW_J: U12 pin 4.

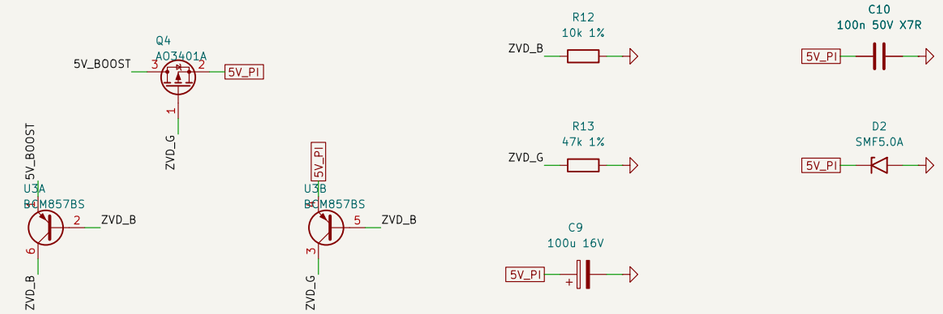
STAT pushes to VDD (5 V): never straight to a GPIO.
Charging: STAT low → D4 on, Q3 off → CHG_STAT = 1.
Complete: STAT high → Q3 on → CHG_STAT = 0. No USB: STAT Hi-Z,
R6 holds Q3 off → CHG_STAT = 1. (USB with no cell / shutdown: Hi-Z,
R5-D4-R6 bias Q3 on → reads 0 and D4 glows faintly.)

## 4 5.1 V BOOST (TPS61022, 1 MHz, auto-PFM)



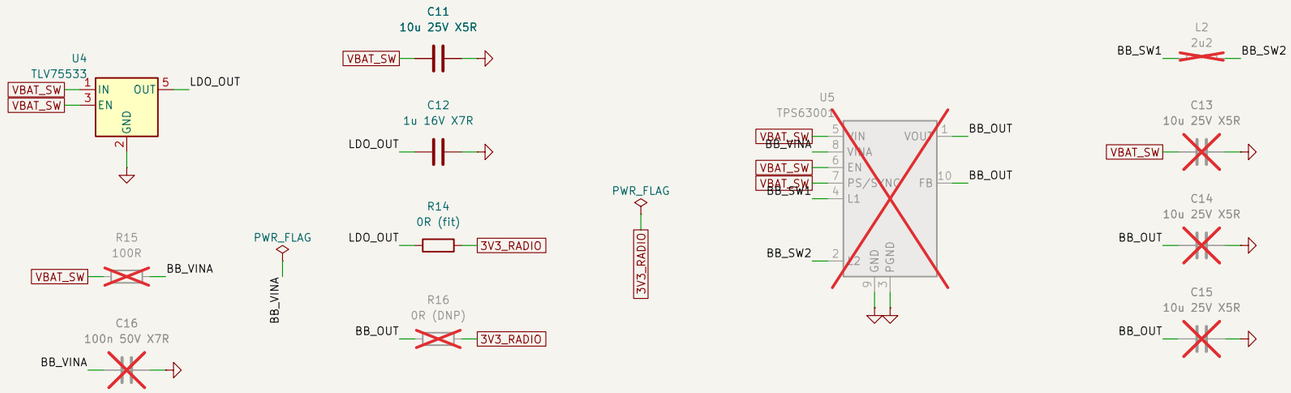

VOUT = 0.600 V x (1 + 750k/100k) = 5.10 V (PFM 5.15 V; OVP 5.5 V min).
L1 744316100 is in TI's Table 8-2: peak ~5.7 A at 3.0 V in / 2.5 A out.
C3 100 pF: feed-forward zero 1/(2π·750k·100p) = 2.1 kHz (§8.2.2.4, COUT_eff
> 40 µF). EN = VBAT_SW: on with the switch. MODE = VIN: forced PWM (layout, 19-Oct: MODE sits
between VIN and EN, so it shares their copper; PFM would only save a few mA below ~100 mA). C5 bulk: the cell sits on a lead
(§8.2.2.5). Tight SW loop, VIN/VOUT caps at the pins; far corner from the SP4.

## 5 BACK-POWER IDEAL DIODE (Pi 3B circuit) & 5V_PI AT THE HEADER



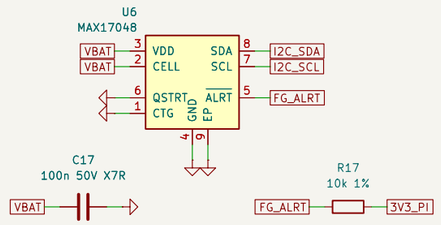

U3A, diode-connected with its emitter on 5V_BOOST, sets ~0.45 mA through R12; U3B mirrors it
against 5V_PI. Forward (5V_BOOST above 5V_PI by more than ~40 mV): U3B starves, R13 pulls ZVD_G
low → Q4 on (≈ 0.15 V at 2.5 A). Back-power (micro-USB on the Pi, or the boost off): 5V_PI >
5V_BOOST, U3B saturates, ZVD_G ≈ 5V_PI → Q4 off, and its body diode (D → S) is reverse biased.
Pi 3B values: 10 k on the mirror base (R12), 47 k on the gate (R13). The Pi 3 A+ has no such
circuit and header pins 2/4 bypass its polyfuse. C9/C10/D2 at header pins 2/4.
Q4: AO3401A (or DMG2305UX as in the Pi). U3: one BCM857BS, matched pair (units A and B; as on the Pi 3B).

## 6 RADIO SUPPLY: TLV75533 LDO (fitted) | TPS63001 buck-boost option (all DNP)



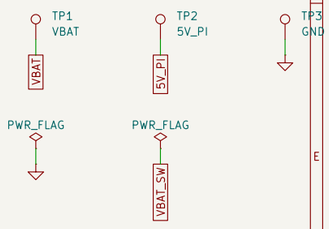

LDO: ≈ 65 mV dropout at 150 mA; COUT 1-200 µF (C12 + the 110 µF at the SP4, radio sheet).
Fit exactly ONE of R14 (LDO, default) / R16 (buck-boost). Buck-boost only if TO shows > 2 dB less
output at a 3.0 V cell: FB = VOUT (fixed 3.3 V), PS/SYNC high = forced PWM (low ripple at the radio).

## 7 FUEL GAUGE (MAX17048, I2C 0x36)




VDD/CELL: Kelvin trace from Q1's source (VBAT), GND to J20, not along the boost input path;
it reads the cell ~I x RDS(Q1) low under load (safe side for the thresholds).
No I2C pull-ups here (the Pi has 1.8 k). Software: never set MODE.EnSleep.

## 8 TEST POINTS




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**MTP F26 Team C (UPC)**

Sheet: /power/
File: power.kicad_sch

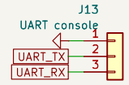
**Title: Power: cell, charger, boost, ideal diode, radio LDO, fuel gauge (DRAFT**

| Size: A3 | Date: 2026-10-19 | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/5 |

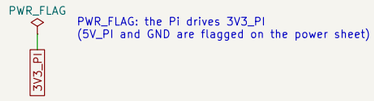
# PI HEADER - Raspberry Pi 3 Model A+ only (65 x 56 mm)

DRAFT - for ET review. The HAT sits on the Pi's 40-pin header (J1, socket on the HAT's underside).
No ID EEPROM: pins 27/28 (ID_SD/ID_SC) stay unconnected (allowed for a non-'HAT' add-on board).
The Pi's WiFi/BT chip is disabled in software or desoldered (rules p.6, Atenea Q22).
5V_PI comes from the power sheet's ideal diode (back-feeds the Pi's 5 V after its polyfuse).
3V3_PI = the Pi's own 3.3 V from pin 17 (pin 1 unused): pull-ups only, no load. The radio has its own LDO (3V3_RADIO).

## UART CONSOLE




1 GND, 2 TX (Pi GPIO 14 out), 3 RX (GPIO 15 in).
Matches common USB-UART leads (their RX to pin 2).
3.3 V levels only: never a 5 V adapter.



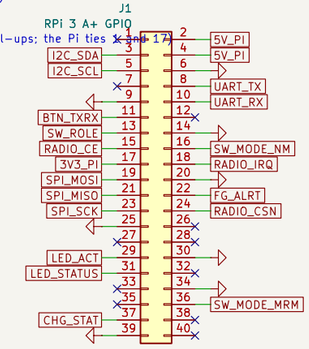

PWR_FLAG: the Pi drives 3V3_PI
(5V_PI and GND are flagged on the power sheet)

## MOUNTING (HAT spec: M2.5, non-plated, not grounded)




Holes at the Pi's 49 x 58 mm pattern (board template places them).
Nylon standoffs: the hole is NPTH, keep copper 3.1 mm away (Ø6.2).

| Header pin | BCM function (pinout.xyz) |
|---|---|
| 1 | 3V3 (unused: pin 17 feeds the pull-ups; the Pi ties 1 and 17) |
| 3 | GPIO 2 SDA1 |
| 5 | GPIO 3 SCL1 |
| 7 | GPIO 4 |
| 9 | GND |
| 11 | GPIO 17 |
| 13 | GPIO 27 |
| 15 | GPIO 22 |
| 17 | 3V3 |
| 19 | GPIO 10 MOSI0 |
| 21 | GPIO 9 MISO0 |
| 23 | GPIO 11 SCLK0 |
| 25 | GND |
| 27 | ID_SD (GPIO 0): keep NC |
| 29 | GPIO 5 |
| 31 | GPIO 6 |
| 33 | GPIO 13 |
| 35 | GPIO 19 |
| 37 | GPIO 26 |
| 39 | GND |




| Header pin | BCM function |
|---|---|
| 2 | 5V |
| 4 | 5V |
| 6 | GND |
| 8 | GPIO 14 TXD0 |
| 10 | GPIO 15 RXD0 |
| 12 | GPIO 18 |
| 14 | GND |
| 16 | GPIO 23 |
| 18 | GPIO 24 |
| 20 | GND |
| 22 | GPIO 25 |
| 24 | GPIO 8 CE0 |
| 26 | GPIO 7 CE1 |
| 28 | ID_SC (GPIO 1): keep NC |
| 30 | GND |
| 32 | GPIO 12 |
| 34 | GND |
| 36 | GPIO 16 |
| 38 | GPIO 20 |
| 40 | GPIO 21 |

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**MTP F26 Team C (UPC)**

Sheet: /pi_header/
File: pi_header.kicad_sch

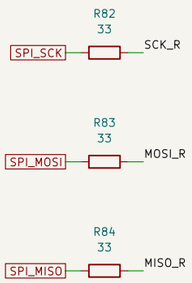
***Title: Pi header: RPi 3 A+ 40-pin, UART, holes (DRAFT)***

| Size: A3 | Date: 2026-10-19 | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/5 |

# RADIO - Ebyte E01-ML01SP4 (nRF24L01P + PA/LNA, 2.4 GHz)

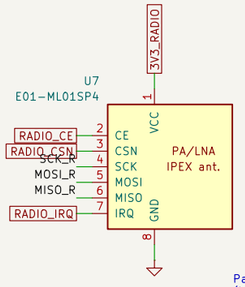
DRAFT - for ET review.
VCC 2.0-3.6 V, ABS. MAX 3.6 V (manual §2.1): only 3V3_RADIO from the TLV75533 LDO on the power sheet, never the cell (4.2 V) and never the Pi's 3V3 (shared with the SoC, TX bursts of 120 mA).
Layout (manual §4.1): GND pour under the module on both layers, stitched; no top-layer tracks under it; keep it away from the boost converter and its inductor (far corner of the HAT); IPEX end toward the board edge.
Antenna: Ebyte TX2400-FPC-2509 flex, off-board on a box wall, on the module's IPEX (U.FL); glue the U.FL plug (drop test).

## SPI SOURCE TERMINATION (33 Ω)



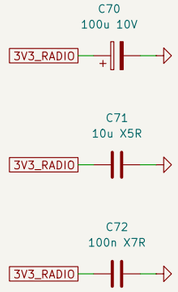

R82 SCK, R83 MOSI: at the Pi end (header J1, drivers = Pi).
R84 MISO: at the radio end (driver = SP4 pad 6).
Source termination of the 30-40 mm lines: cheap ringing control, 0 Ω if the bench shows no need (spidev runs at ≤ 8 MHz).

## MODULE



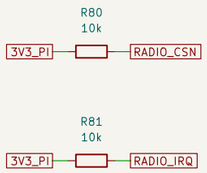

Pads 9/10 = GND half-holes on the long edges (hidden, stacked on pin 8). CE also enables the LNA (manual §4.2): driven by GPIO 22, whose boot default pull-down keeps the radio idle until the driver starts.

## DECOUPLING AT PAD 1 (110 µF total; TLV75533 allows 1-200 µF)




Place all three at pad 1, smallest closest. C70 rides the 120 mA TX bursts (PA on) so the LDO's load step does not dip VCC.

## PULL-UPS TO 3V3_PI




R80: CSN high = radio deselected while the Pi boots (GPIO 8 is an input with its default pull-up until spidev takes it).
R81: IRQ idle level while the SP4 is unpowered or in reset; harmless otherwise.
Both rails come from VBAT_SW together (boost → Pi, LDO → radio): no back-feed of the Pi.

## TEST POINT




At the module: TO checks VCC under TX bursts (3.6/3.3/3.0 V cell).

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DRAFT - for ET review
**MTP F26 Team C (UPC)**

Sheet: /radio/
File: radio.kicad_sch

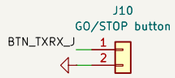
***Title: Radio: E01-ML01SP4 (DRAFT - for ET review)***

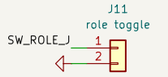
| Size: A3 | Date: 2026-10-19 | **Rev: DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/5 |

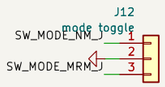
# UI - panel controls on JST-XH leads, bench button, LEDs

DRAFT - for ET review. Software: proto/port/linux/ui.c (inputs active low with the Pi's pull-ups; LEDs active high).
Mode switch (J12, ON-OFF-ON): GPIO 23 low = NM, else GPIO 16 low = MRM, else (centre) SRI. Role (J11): low = TX.
External 10 kΩ pull-ups: the BCM2837 boot default on GPIO 16/17/23/27 is a PULL-DOWN, so an open toggle
would read 'active' until Linux sets the pull-up. 1 kΩ series: GPIO 16 may be driven by legacy firmware at boot
(HAT design guide); 1 kΩ limits that to 3.3 mA into a closed switch. ESD array right at the connectors.
Panel holes: Ø7.5 mm for the PS1024 button, Ø6.5 mm for the 100-series toggles.
Glue the XH housings to the board and the leads to the box (drop test).

## CONNECTORS (panel parts off-board)




J10: panel button E-Switch PS1024ABLK (momentary).
Bench: plug the panel button into J10 (or bridge its two pins); no on-board button.
J11: role toggle E-Switch 100SP1T1B1M1QEH, closed = TX;
  may stay unfitted until Atenea Q8 (open = RX).
J12: mode toggle E-Switch 100SP3T1B1M1QEH ON-OFF-ON:
  1 NM, 2 common (GND), 3 MRM; centre = SRI.

## PULL-UP 10 kΩ (connector side)













## SERIES 1 kΩ





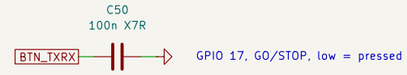


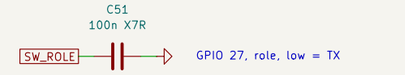


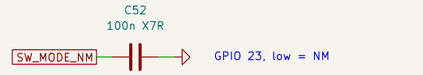




## 100 nF AT THE GPIO



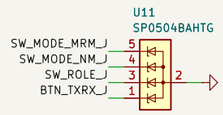







## ESD (at the connectors)




4-line TVS array, 5.5 V standoff, 30 pF, ≤ 100 nA leakage
(datasheet p.2), IEC 61000-4-2 20 kV contact.
Pin 2 = common → GND; 1/3/4/5 = lines (datasheet p.1).

## LEDs (active high, ≈ 4 mA from 3.3 V)




D10 activity (GPIO 5), D11 status (GPIO 6).
(3.3 − −2.0 V) / 330 Ω ≈ 4 mA, well under the
Pi's 16 mA per pin. No InGaN (blue/white: Vf too high).

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**MTP F26 Team C (UPC)**

| | | |
|---|---|---|
| Sheet: /ui/ | | |
| File: ui.kicad_sch | | |
| ***Title: UI: panel button, toggles, LEDs (DRAFT - for ET review)*** | | |
| Size: A3 | Date: 2026-10-19 | **Rev: DRAFT** |
| KiCad E.D.A. 10.0.6 | | Id: 5/5 |